

Pradeep Yenkuwale

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PROFESSIONAL SUMMARY

Senior software engineer (8+ years) specialising in **AWS-deployed distributed systems, JavaScript/TypeScript backend services, and event-driven microservices (Node.js, NestJS)**. Architected production systems on **AWS (EC2, S3, Lambda, IAM, CloudWatch)** serving 10K+ transactions/day at Almatar; eliminated production crash class via MongoDB re-architecture (40% CPU reduction). Hands-on with **Python (FastAPI, Flask)**, GenAI integration patterns, and prompt engineering. Solo-architected and deployed Zyntra - a real-time social platform on AWS - owning full-stack decisions from schema to production. Strengths in distributed systems, AWS architecture, idempotent service design, and cross-team integrations.

TECHNICAL SKILLS

Cloud & AWS: AWS EC2, S3, Lambda, IAM, CloudWatch, API Gateway, SQS, SNS, RDS, DynamoDB, ECS, VPC, CloudFormation

AI / GenAI: LLM Integration (OpenAI, Anthropic Claude, AWS Bedrock), Prompt Engineering, RAG Architecture, Vector Embeddings, Function Calling.

Languages & Frameworks: JavaScript, TypeScript, Node.js, NestJS, Express.js, Flutter (Dart), Angular, React, Python (FastAPI, Flask).

Databases & Caching: MongoDB, MySQL, SQL Server, IBM DB2, Redis, DynamoDB, Drift, Data Modelling, Query Optimisation, Indexing.

Messaging & Real-Time: Apache Kafka, RabbitMQ, AWS SQS/SNS, WebSockets (Socket.IO), Server-Sent Events (SSE), Event-Driven Architecture

Architecture & Design: Microservices, HLD/LLD, Distributed Systems, REST API Design, Cache-First Architecture, SOLID, Design Patterns (Adapter, Guard, Strategy)

DevOps & Observability: Docker, GitLab CI, AWS CloudWatch, Grafana, CI/CD, OAuth 2.0, OIDC, JWT, Logging, Monitoring, Metrics, SLI/SLO

INDEPENDENT PRODUCT - ARCHITECTURE & ENGINEERING

Zyntra - Social & Real-Time Messaging Platform

2026–Present

Consumer social platform - 1:1 chat, social feed, matchmaking, relationships. Solo-architected end-to-end; deployed on AWS; pre-public-launch security closeout phase.

Stack: AWS (EC2, S3, IAM, CloudWatch), NestJS, TypeScript, MongoDB, Redis, Apache Kafka, WebSockets (Socket.IO), Flutter (Drift), Docker

- **AWS-deployed microservices architecture** – 5 services (auth, user, messaging, matchmaking, notification) on EC2 with Docker, communicating via Kafka event contracts for asynchronous, decoupled workflows. CloudWatch for logs/metrics; IAM policies for least-privilege service access.
- **Multi-state auth + tier-capability framework** - three-stage user journey (temporary -> registered -> verified) with stateless tier claims in JWT; built a metadata-driven guard system using NestJS decorators (@RequiresCapability, @OwnerBypass) to enforce feature access declaratively without per-controller boilerplate.
- **Multi-tier API throttling** - custom hybrid guard with dual short/long buckets and per-user/per-IP keying; propagated upstream 429s across 4 consumer services so user-facing errors stay clean; suppressed rate-limit headers (Retry-After kept) to block throttle-config recon.
- **Cache-first Flutter data layer** - Drift-backed local store with on-demand sync and server reconciliation; feed/profile/messages render offline-first with background convergence that never blocks the UI.

- **Secure AWS S3 media pipeline** - direct-to-S3 uploads via time-scoped pre-signed URLs (server-side token issuance, IAM-scoped credentials), metadata reconciliation post-confirmation.
- **Multi-level privacy controls** - enforced user-scope and conversation-scope privacy policies at both the query layer and real-time event-dispatch layer to prevent leaks through WebSocket channels.
- **Designing image-based community moderation** - Claude (Anthropic) integration for multimodal NSFW classification on user-uploaded media; verdicts to route through the Trust & Safety enforcement pipeline.

PROFESSIONAL EXPERIENCE

Senior Software Engineer - Full Stack | Almatar Software Solution Pvt Ltd

Mar 2024 – Present

Online Travel Agency (OTA) serving 10K+ bookings/day: flight booking, fare search, recommendations, and Add-on services.

- **Own 4 backend microservices on AWS** across booking, fare search, recommendations, and airline integrations - Node.js, NestJS, Python utilities, MongoDB, Redis, Apache Kafka, RabbitMQ, REST APIs; Authored 5+ design documents adopted as squad-level patterns; shipped to production, own on-call.
- **Led re-architecture of a MongoDB bottleneck** where a single document aggregated all supplier/airline data, driving unbounded growth and service crashes; isolated data into per-supplier/airline documents with a backwards-compatible read/write path; coordinated zero-downtime migration on AWS infrastructure - **40% lower CPU, 25% lower memory, crash class eliminated**, with downstream AWS compute cost reduction through right-sized instances.
- **Designed and led the branded fare recommendation engine** - architecture for ranking fare families by traveller intent and business rules; defined feature contracts, ranking signals, and feedback loops - **20% engagement lift, 20% growth in add-on sales**. Exploring LLM-augmented ranking for the next iteration.
- **Led cross-team integrations with Ads and User Loyalty platforms** - aligned with 2 platform teams, versioned API contracts, and established the event-driven integration pattern adopted by adjacent booking-flow features.
- **Partnered with the Centre of Excellence (COE) on observability and performance** - built AWS CloudWatch + Grafana dashboards, API health monitors, and alerting against SLO budgets; drove perf fixes - **20% p95 latency improvement** and reduced AWS compute costs through profiling-driven optimisation.
- **Built a multi-modal chatbot system for customer-support** automation (Python, FastAPI, LangChain + OpenAI) - text-to-text, voice-to-text, and voice-to-voice; abstracted prompt templates and model providers behind a unified interface for prompt/A-B iteration.
- **Mentored 3-4 junior engineers** (service architecture, code reviews, code quality) and drove the technical hiring loop - interviewed 10-15 backend candidates and contributed to levelling decisions.

Senior Software Developer - Full Stack | IBM India Pvt Ltd

Jun 2022 – Feb 2024

Enterprise telecom automation - AT&T DCP-based international toll-free number provisioning across 60+ countries.

- **Built Node.js + Python microservices** from the ground up with reusable architecture, secure REST API authentication, and cloud-deployable design - **50% automation of provisioning, 90% less manual effort, 70% lower operational cost** across 60+ countries.
- **Improved automation pipeline latency by 20%** via service-level optimisations and streamlined integration contracts with AT&T DCP and carrier back-office systems.
- **Technically led a team of 4 junior engineers** - drove architecture and code reviews, task delegation, and mentoring on microservice design, API contracts, and security-by-default; 100% on-time delivery.
- **Owned cloud deployment and container hardening** (Docker, CI/CD) for high-traffic enterprise workloads; implemented IAM-style security protocols and drove security reviews - **30% reduction in vulnerabilities** aligned with enterprise compliance standards.

Software Engineer - Full Stack | Irisidea Tech Solutions Pvt Ltd

Feb 2021 – May 2022

Client engagement – Boons (US hyper-local food delivery, multi-provider fallback dispatch).

- **Designed and built an internal NPM package for multi-provider fallback delivery routing** using the adapter pattern over DoorDash, Uber Eats, and Postmates - normalised divergent auth, contracts, and webhook formats

behind a single interface; engineered idempotent dispatch and retry handling to prevent double-shipments under transient failures. Pattern directly reusable for multi-LLM provider abstraction (OpenAI / Bedrock / Anthropic fallback).

- **Built Clover POS integration microservices** (Node.js, Express.js, MongoDB) - order acceptance, terminal creation, lifecycle tracking, and payment-capture reconciliation (90% unit test coverage, 50% fewer post-deploy defects); decoupled scheduled jobs into a standalone Node.js scheduler to isolate load from the main API.

Software Developer Intern → Project Engineer | Solize India Technologies Pvt Ltd Jan 2017 – Jan 2021

Automotive engineering services – internal tools + platforms for design pipelines (client: Tenneco, Germany).

- **Built Python desktop tools and automation scripts** (data processing, file pipelines, batch backups) for automotive client Tenneco - 50% effort reduction, 80% less manual maintenance, standardised outputs for downstream design pipelines.
- **Developed an internal full-stack platform** (Node.js + Angular) with REST APIs for engineering workflows; optimised legacy codebase for 30% faster execution on long-running data jobs.

EDUCATION

Master of Computer Applications (MCA) - Visvesvaraya Technological University, Bengaluru 2014 – 2017

B.Sc., Computer Science - Gulbarga University, Gulbarga 2011 – 2014

CERTIFICATIONS & AWARDS

- **Client & Partner Success Award - IBM India (2022)** - led AT&T DCP automation across 60+ countries (90% manual effort reduction, 70% cost savings).
- **Performance Excellence Award - IBM India (2022)** - 50% process automation and 100% on-time delivery on the global voice services program.